Identifying and evaluating algebraic expressions

1 Match the following terms with their definitions:

a Constant	**i** Symbol that represents an unknown number
b Variable	**ii** A purely numeric term in an algebraic expression
c Coefficient	**iii** The value that indicates how many of a variable in a term
d Algebraic term	**iv** Term including a variable
e Algebraic expression	**v** A collection of variables, numbers, grouping symbols, and operations.

2 State the numerical coefficient of $-2.58t^4$.

3 Consider the expression $x + 6$.

- **a** Find the value when $x = 4$.
- **b** Find the value when $x = 9$.
- **c** What happens to the value of $x + 6$ as we substitute in different values of x?

4 Evaluate the following expressions to one decimal place:

- **a** $14.5 - 4x$ when $x = 4.2$
- **b** $18.6 - 3x$ when $x = 4.1$

5 If $m = -3$ and $n = 4$, evaluate the following expressions:

- **a** $mn - (m - n)$
- **b** $m^2 + 9n$

6 Evaluate the following expressions:

a $\dfrac{m}{2m}$ when $m=5$

b $\dfrac{6}{7x+3}$ when $x=3$

c $\dfrac{a}{11}+\dfrac{b}{11}$ when $a=5$ and $b=2$

d $\dfrac{a}{1-r}$ when $a=3$ and $r=\dfrac{1}{10}$

e $\dfrac{p+4}{q}$ when $p=-8$ and $q=-8$

f $\dfrac{3a}{2b-6}$ when $a=4$ and $b=14$

g $\dfrac{2p+4}{3q+5}$ when $p=2$ and $q=4$

h $\dfrac{4r-2}{3s-5}$ when $r=6$ and $s=7$

i $\dfrac{x^2}{3}+\dfrac{y^3}{2}$ when $x=-4$ and $y=3$

j $s\left(12-t\right)$ when $s=-3$ and $t=10$

k $x\left(2-4y^2\right)$ when $x=-2$ and $y=-2$

l $a\left(b-13\right)+b^2$ when $a=-9$ and $b=10$

m $3a^2-4b^3$ when $a=4$ and $b=2$

n $8p^2+5q^3$ when $p=3$ and $q=-2$

o $5y-z^2+9$ when $y=7$ and $z=4$

p $\dfrac{1}{2}y^2-6z+1$ when $y=-2$ and $z=\dfrac{5}{6}$

q $y^3+4yz-2x^2$ when $x=-4$, $y=2$ and $z=1$

r $\left(u+v\right)\left(w-y\right)$ when $u=5$, $v=8$, $w=2$ and $y=10$

7 For $x=5$ and $y=4$, evaluate $\sqrt{2x^2+4y+6}$ correct to two decimal places.

Additional questions

8 Write the expression $9x$ in words.

9 State whether the following expressions are [redacted] terms.

a $11p$ and $3p$ b $2p$ and 15 c $9p$ and $6q$ d 12 and 8

e $5h$ and $5hk$ f $4h^4$ and $3h$ g $8h^3k^4$ and $9k^4h^3$ h $6h^k$ and $7k^h$

10 For the following algebraic expressions:

i State the number of terms.

ii Identify the numerical coefficient of the first term.

iii Identify the constant term.

iv State whether it contains like terms.

a $2x+4$ b $7y+3+5x$

c $3x+2y-8x+9$ d $8p+5$

11 Consider the algebraic expression $3x+10+4x$.

a State the number of terms.

b Identify the numerical coefficient of the third term.

c Identify the constant term.

d State whether it contains like terms.

12 For each of the following, determine if the statement is true or false. Explain your reasoning.

a An expression must include a variable.

b Any expression with more than one term can be simplified.